E-WASTE GOVERNANCE MODEL: NATIONAL VERSION FOR INDIA

Collect, pay, securely erase, sort, recycle and educate: design review and implementation route at national scale

The uploaded Thapar campus model, redesigned for India as a whole and checked against official data and law

Based on	E-Waste Governance Model and Implementation Route (uploaded PDF, prepared 12 October 2026), kept in the same structure
Scope change	From one campus pilot to a national system across states, with a campus pilot retained as phase 0
Date	20 October 2026
Group / roll numbers / names	_____
Companion document	E-Waste_Platform_India_Final_Specification.pdf (website build specification)
Status	Planning document, not legal, tax or financial advice. Items marked CONFIRM need written confirmation.

Reading note

Facts carry a source number such as [N2]; the list is at the end. Official sources come first; summaries by law firms and compliance sites are marked secondary and should be checked against the gazette text.

Items marked [inference] or [design] are my reasoning or recommendations, not facts. Money figures are symbolic because real prices must come from local quotes.

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What you proposed, restated for India

- Collect e-waste from consumers and companies across India and pay them a modest amount for each item.
- Run every device through a data-erasure pipeline: collection point, serial and IMEI scan, wiping, automated verification, audit record; dead devices go to physical destruction.
- Send e-waste only to designated, CPCB-registered recyclers, where computer vision helps segregate products.
- Run national awareness campaigns on reduce and reuse.

Verdict in short

- **The core idea is still sound, but the national picture has changed.** The first PDF said India formally recycled about a third of its e-waste. Official CPCB figures now show 61.94% in FY 2023-24 and 70.71% in FY 2024-25 [N1], and the Ministry reports 9,79,080 tonnes recycled out of 14,14,645 tonnes generated in FY 2025-26 [N2]. So the case for a national platform no longer rests on the claim that most waste misses recyclers. It rests on four things the official numbers do not show: trust in data erasure, reuse, traceability of what actually reaches recyclers, and access for households and small institutions.
- **The official percentage is modelled.** CPCB estimates generation from producers' sales and average product life, and processed quantities come from registered entities [N1]. A national platform that records real devices, weights and receipts would supply independent ground truth, which is its most defensible national value.
- **The weakest link is still payment.** Informal buyers are reported to pay at least double what formal recyclers offer (first PDF). A modest payment cannot outbid them. Compete on a data-erasure certificate, convenience, reuse value and institutional demand.
- **Computer vision belongs at the recycler.** At hubs use it for intake classification only.
- **Scale in phases with gates.** Campus pilot, city, state, region, nation; each step needs measured results and legal sign-off (section 12).

PART 1. The national governance model

1. Who does what

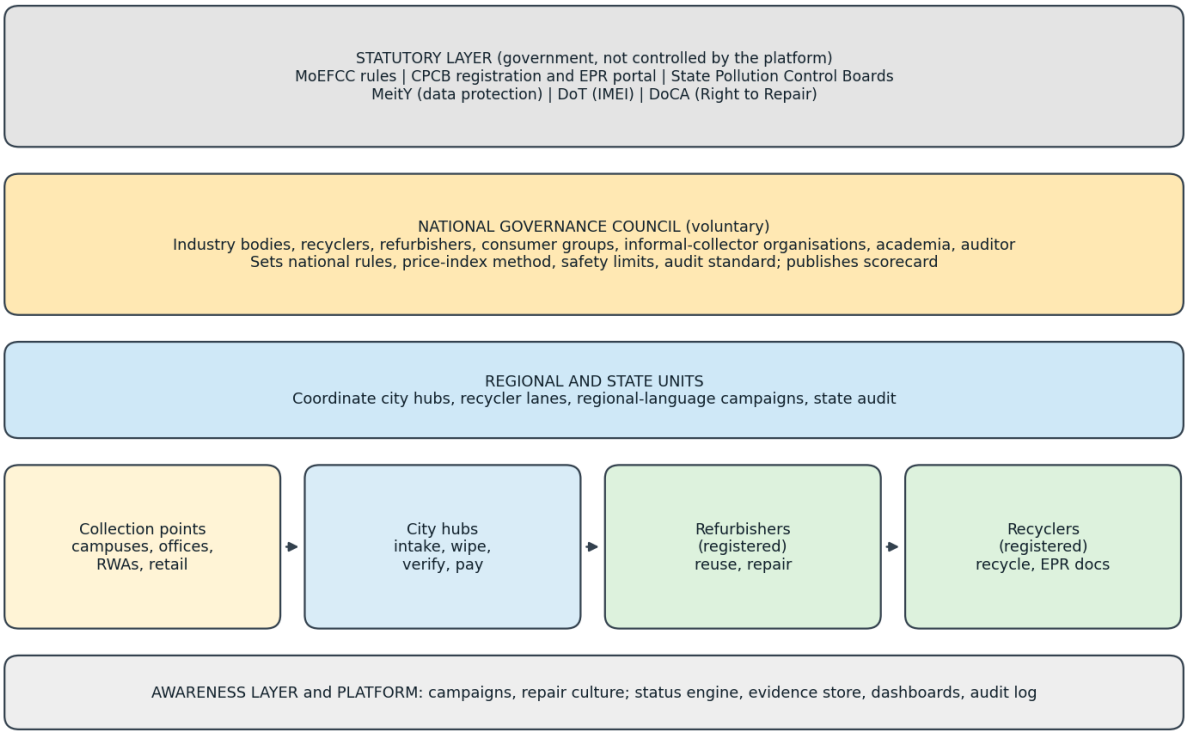


Figure 1. Devices flow left to right through the lower row. Money flows right to left: material sales, reuse income, paid disposal and approved EPR income fund collection, wiping and payouts. Data moves between levels only as signed certificates and minimum records.

Actor	Role	Key rules
Statutory layer (government)	MoEFCC makes the rules; CPCB runs registration and the EPR portal; SPCBs enforce locally; MeitY, DoT and DoCA cover data, IMEI and repair [N3, N4, N9, N11, N12]	Not controlled by the platform; the platform only complies, reports and cooperates
National Governance Council [design]	Owns the voluntary national rules: accepted items, price-index method, safety limits, data rules, partner admission, audit standard; publishes the national scorecard	Balanced membership including consumer and informal-collector voices; CPCB invited as observer; independent auditor reports to it
Regional and state units [design]	Coordinate hubs, state recycler lanes, regional-language campaigns, state audit, grievance handling	Each unit has a compliance officer and a data protection officer
City hubs (operators)	Run bins, booking, intake, triage, wiping and payouts	Trained staff only; no dismantling; every action logged
Contributors	Households, students, institutions, offices, companies	Sign ownership and consent declaration; accept erasure and payment terms
Registered refurbisher	Repairs and resells or donates reusable devices	Must be registered on the CPCB portal and hold valid consent to operate [N3]
Registered recycler	Dismantles and recycles, using vision and sensor sorting at its plant	CPCB-registered; weighbridge slips, EPR documentation, recovery report
Registered dismantler	Dismantles equipment in line with CPCB guidelines	Since the 2024 amendment a dismantler is defined by compliance with CPCB guidelines rather than by SPCB authorisation alone [N5]
Independent auditor	Samples wiped devices, checks logs, reconciles weights and payments	Separate from operators; reports to the Council

Why a governance model, not just a collection drive

A drive collects once. A governance model fixes who is accountable for each device from drop-off to final recovery, so trust, especially on data, can be earned and repeated. The 2022 Rules place duties on producers, refurbishers and recyclers and require bulk consumers to hand e-waste only to registered entities, but they do not require institutions to count what they discard (first PDF). A voluntary national model supplies that missing accountability and gives the modelled national numbers an independent cross-check.

2. The pipeline, strengthened

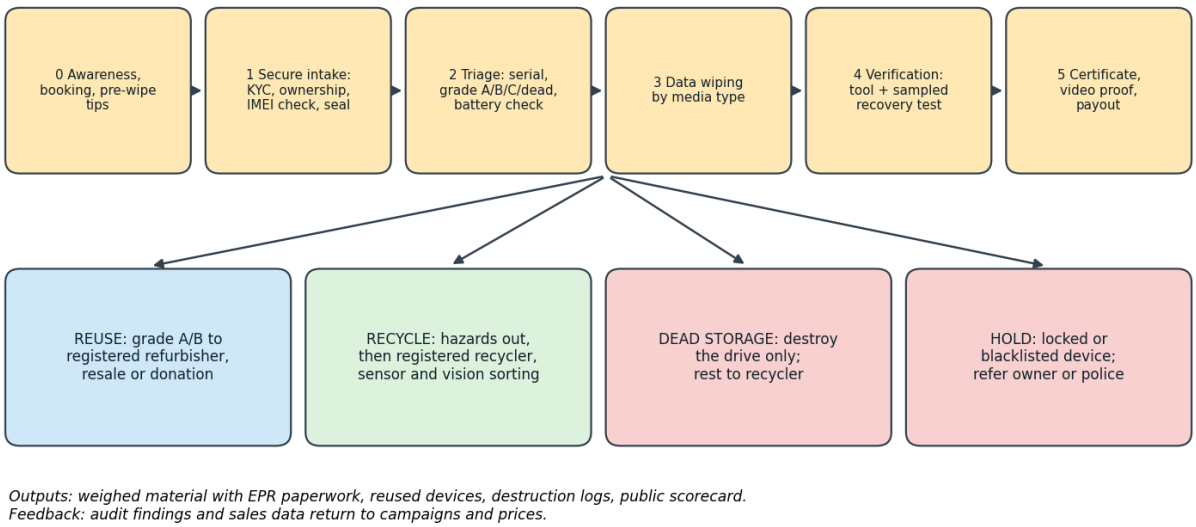


Figure 2. Enhanced pipeline. Amber stages are from your blueprint with the changes described in Part 5; the four paths below are the new branches.

Standards behind the wiping stage

- NIST SP 800-88 Rev. 2 (September 2025) defines Clear (logical overwrite or reset), Purge (overwrite, block erase or cryptographic erase that makes recovery infeasible) and Destroy (physical). It expects the outcome to be validated and documented, and points to IEEE 2883-2022 for the technique for each storage type.
- A factory reset is only Clear. Plain overwriting is unreliable on SSDs and flash because of wear-levelling and over-provisioned blocks, so purge on flash needs the drive's secure-erase, sanitise or cryptographic-erase command. Where data is very sensitive, destroy the storage as well.
- Read the standards themselves before quoting them; the first PDF's sources for them were vendor and commentary sites.

Media	Suggested method (confirm against IEEE 2883)	Verification
Hard disk	Overwrite (Clear) for low sensitivity; purge or destroy for sensitive data	Read-back sampling; log certificate
SATA SSD	ATA Secure Erase or cryptographic erase (Purge); plain overwrite is not enough	Tool status plus sample forensic test
NVMe SSD	NVMe Sanitize (block or crypto erase); a format only clears the namespace	Controller completion plus sample test
Phone or tablet	Remove accounts and activation lock, encrypt, factory reset (Clear); vendor cryptographic erase where available; destroy if it cannot be reset	Boots to setup; sample forensic check
SD card, USB stick	Treat as flash: purge by the device method or destroy	Sample check
Network gear, printers	Reset to defaults, then sanitise internal storage	Configuration check

Media	Suggested method (confirm against IEEE 2883)	Verification
Dead or failed media	Destroy the storage component only; recycle the rest	Photo and weight log

National consistency. A single national wipe procedure, one tool list per media type and one certificate format keep results comparable between hubs. The Council reviews the list yearly [design].

PART 2. Route and ideas to implement nationally

3. Scope: campus pilot first, then widening circles

The first PDF says the full model needs registrations, funding and equipment a student group cannot obtain in one semester. That still holds, so the national model is built outward from the pilot.

Layer	Phase 0: campus pilot (do now)	National model (describe, then build in gated phases)
Collection	Red bins, booking form, hostel and department champions, one collection day a month	Campus, office, residents' association and retail points; scheduled pickup in each city; company contracts; registered informal aggregators as collectors
Payment	Points or certificates first; price survey of local scrap dealers and refurbishers	UPI, IMPS or bank payout via a regulated provider, with KYC, caps and fraud controls; regional price index
Intake and tracking	QR passport per item, spreadsheet, IMEI check, ownership declaration	Web platform with status engine, evidence store and audit log; Sanchar Saathi checks [N11]
Data wiping	Demonstrate on a consented donated device; hold and escort all other devices to the registered recycler	City sanitisation hubs with parallel wiping stations and audited tools
Reuse	Repair and reuse cafe; donation through a registered refurbisher	Refurbisher network; resale channel; link to the Right to Repair portal [N12]
Sorting	Manual sorting by trained volunteers; vision classifier prototype for intake	Vision and sensor sorting at registered recyclers' plants
Awareness	Orientation talk, QR posters, hostel competition, reels, pledge	Multi-year campaign in regional languages
Measurement	Baseline audit, weights, surveys, energy and carbon estimate	National dashboard, annual independent audit, public scorecard

4. Things you can build and demonstrate

Deliverable	What to build	Why it helps
E-Waste Passport	Booking form plus register with auto-generated QR codes and status columns; later the full web platform (see companion document)	Shows tracking and chain of custody
Computer-vision classifier	150-300 original photos for each of 6-8 classes; train a small open-source detector; report precision and recall on held-out images; align class names with the Schedule I categories nationally	Demonstrates the vision step with original data
Data-erasure demo	Wipe a donated drive or phone with consent; show before-and-after recovery test and the certificate	Shows the trust step end to end
Price calculator	Spreadsheet using the pricing formula with real local quotes	Shows honest economics
Energy and carbon sheet	kWh per wipe with a plug-in meter; transport km; avoided emissions from reuse	Ties the project to Energy and Environment
Awareness pack	Poster, 60-second reel, flash cards, pledge card in Hindi, Punjabi and English	Campaign design evidence
Pilot results	Kilograms collected, items tagged, items reused, survey change, reconciliation with recycler receipt	Real data for feasibility

5. Pricing and payment at national scale

Offer to contributor = max(S, a x R) minus H, with a floor and a cap. S is scrap value (weight times the recycler's quoted rate for the category). R is expected resale value after grading. a is the share of R passed to the contributor. H is handling cost per item.

- Build a **regional price index**: each city's rate table is refreshed monthly from at least three recycler quotes and two refurbisher quotes, and the Council publishes the method [design].
- Run the first price survey in the pilot: ten local scrap dealers and two refurbishers, same items. It is also your evidence on competitiveness.

Funding source	How it works	Reliability
Resale of refurbished devices	Margin on grade A and B devices sold or donated at a fee through a registered refurbisher	Medium
Recycler payments for material	Recycler pays for weight or material content	Variable with metal prices
EPR-linked income	Partner producers or recyclers pay for collected quantities (certificates)	Uncertain; a 2026 field study argues certificates reward paperwork more than collection
Institutional budget or CSR	Sustainability budgets or corporate CSR funds	Good for pilots
Paid secure disposal for companies	Companies pay for certified disposal (IT asset disposition)	Strong; companies value the certificate
Deposit-refund style funding	A national scheme paying for returned devices, as suggested in the first PDF's policy references	Policy dependent; not in force as far as I found (CONFIRM)

Controls that prevent abuse

- KYC and bank or UPI payout to the verified owner only; per-person and per-day caps.
- Ownership declaration, and for phones an IMEI check on Sanchar Saathi, whose Know Your Mobile function lets anyone check whether an IMEI is genuine [N11]. Blacklisted, duplicate or blocked IMEIs are not bought and are referred to the owner or police.
- No cash for items of unclear origin; photograph at intake; keep logs for audit.
- Do not collect or store Aadhaar numbers unless a law requires it [design].

Competing with the informal market

The first PDF cites a policy study saying unauthorised recyclers handle most e-waste because they offer at least double the prices, and a 2026 Delhi field study describing cash-paying collectors outbidding formal recyclers because they carry no compliance cost. A national platform should therefore: pay for reuse above scrap; sell trust (certificate, privacy, convenience) to institutions that cannot lawfully use unregistered buyers; and register informal aggregators so they are partners, not opponents. The Battery Rules already allow informal collectors to register on the portal [N8].

PART 3. Sorting with computer vision, and national awareness

6. Computer vision: what it can and cannot do

- **How it works.** Cameras on a conveyor feed a deep-learning model; a robot arm with suction grippers or air jets moves each item to the right bin; near-infrared sensors add material information.
- **Evidence.** A leading vendor reports about 80 picks a minute, up to about 150 with tandem units, and over 160 systems installed worldwide (company claims, per the first PDF). An independent 2026 university proof of concept reached about 81.8% accuracy in controlled tests.
- **Limits [inference].** It sees only the outside, so it cannot detect a swollen battery inside a laptop; clutter and dirt reduce accuracy; new designs need retraining; installation is justified only by large volumes.
- **Where it belongs.** At the registered recycler's plant. For hubs, use a phone-camera classifier that suggests category and weight class at intake, with a person responsible for hazard checks.
- **National step.** Agree one category list, mapped to the 106 equipment categories in Schedule I of the Rules [N3], so hub intake codes and recycler records match.

7. National awareness that changes behaviour

Campaigns should answer three questions: where do I take it, is my data safe, and what do I get. Surveys cited in the first PDF found low policy awareness (only 6.5% of surveyed engineering students knew the policy), hoarding and unclear drop-off points.

Element	National design	Measure
Launch	Multilingual posters, SMS and QR codes stating the nearest drop-off and the data promise; partner with residents' associations, universities, schools and offices	QR scans, posters seen
Competition	City, institution and hostel leaderboards for kilograms and items reused; certificates	Items per participant group
Trust message	Open 'see your device wiped' sessions; show a sample certificate; public verification page	Survey of trust in the process
Repair and reuse	Monthly repair cafes with mentors; point people to the Right to Repair portal for manuals, spare-part information and service centres [N12]	Devices repaired
Reduce first	Short guides on extending device life, buying repairable products, avoiding trade-in churn	Purchase-intent survey
Storytelling	Student and creator reels on where e-waste goes and informal-recycling health risks, in regional languages	Reach and shares
Feedback	Monthly national and state scorecards	Participation trend

Reduce and reuse before recycling. Studies cited in the first PDF say manufacturing dominates a device's footprint (for a laptop 62-70% of life-cycle energy), so keeping devices in use saves more than recovering metals. The Right to Repair framework was set up by the Department of Consumer Affairs for mobile phones and tablets, consumer durables, farm equipment and automobiles, and the portal gives repair information [N12].

PART 4. Advantages and challenges the system addresses

Challenge today	How the system helps	Evidence
People hoard old devices	Convenient drop-off, a visible payoff, a trusted wipe	Surveys cited in the first PDF find storing is the commonest action; data worries and not knowing where to recycle increase it
Fear of data leaks	Certified, standards-based erasure with a per-device certificate	NIST SP 800-88 Rev. 2 and IEEE 2883 give a recognised basis
Modelled national data	Device-level register and weighed outputs give an independent check	CPCB figures derive from producer sales and product life [N1]
Waste outside traceable chains	Registered partners, paperwork and geotagged dispatch evidence	Reports of fake certificates and collected waste sold to unauthorised recyclers (first PDF)
Uneven state collection	Regional units and recycler lanes; hubs placed where collection is weak	Maharashtra, Rajasthan and Punjab collect less relative to size [N13]
Reusable devices scrapped	Grading and a refurbisher branch	Rules recognise refurbishing certificates [N3, N6]
Hazard and fire risk	Battery check at triage; sealed battery stream; registered battery recyclers	US EPA recorded 245 lithium-battery fires at 64 waste facilities, 2013-2020 (first PDF); Battery Rules apply [N8]
Low awareness	Embedded multilingual campaigns with feedback	6.5% policy awareness among surveyed engineering students
Slow manual sorting at recyclers	Vision-guided sorting at plant scale	Vendor claims; limited independent evidence
Businesses need proof of secure disposal	Auditable chain of custody and certificates	Supports data-protection duties under the DPDP Act [N9]

Additional benefits for Energy and Environment

- Avoided mining and manufacturing energy through reuse and recovery.
- Lower exposure for informal workers when material moves to registered recyclers.
- Data for campus and state sustainability reports and institutional rankings.
- A base for solar-powered hubs, linking to renewable-energy coursework.

PART 5. Flaws in the original flow and how to fix them, updated for India

#	Flaw in the blueprint	Why it matters	Fix
1	Wiping only after devices reach the hub	Data exposed in transit and storage	Owner pre-wipe tips; tamper-evident bags; encryption; logged custody; secure cage
2	No ownership or legality check	Stolen or blocked phones bought; fraud	Ownership declaration, ID, IMEI check on Sanchar Saathi, refuse blocked devices
3	Locked devices not handled	Cannot be wiped or reused	Owner signs out before drop-off; hold locked devices and refer to the owner
4	'If dead, destroy' treats the whole device	Wastes recoverable material	Destroy only the storage; recycle the rest
5	No reuse branch	Working devices scrapped	Grade at triage; refurbish through a registered refurbisher
6	Verification only by software	A tool can report success without proof	Sampled forensic recovery tests by an independent person
7	Audit cloud may hold personal data	Breach target; DPDP duties	Minimal data (device ID, method, time, result); sign reports; set retention; host in India
8	No hazard stage	Fires and releases	Remove batteries and lamps at triage into a sealed, labelled stream
9	Payment stage missing	Fraud, disputes, legal exposure	Pricing formula, KYC, caps, receipts, grievance process
10	Registration and legality unspecified	Dealing with unregistered entities is not allowed for registered ones	Partner only with registered refurbishers and recyclers; check the CPCB portal; legal advice before reselling
11	'Collectors send to designated stations' without traceability	Documented sale of collected waste to unauthorised recyclers	Weighbridge slips, manifests, random audits, geotagged handover logs
12	Vision at the recycler but unmatched to the hub	Mismatch of categories and data	Shared category codes mapped to Schedule I; vision at intake for classification only
13	Awareness is the last step	Demand and trust come first	Start before launch; keep running with feedback
14	Trade-in incentives may encourage upgrading	Buy-back can shorten device life [inference]	Pair payment with repair-first messages; pay more for reuse than scrap
15	National: single operator and no oversight	Concentration and conflicts of interest	Governance Council with balanced membership and an independent auditor
16	National: one national price	Local prices differ	Regional price index refreshed monthly
17	National: long-distance transport to recyclers	Cost, emissions, leakage risk	State lanes, consolidated trips, geotag and weight reconciliation
18	National: language barriers	Exclusion; consent not informed	Regional-language notices and consent text
19	National: storage limits ignored	Legal exposure	Alert at 150 days; storage limit stated as 180 days in secondary summaries (CONFIRM) [N7]

PART 6. Limitations of today's system that the model tries to improve

Limitation	Detail and evidence
Formal processing still incomplete	CPCB reports 70.71% collected, dismantled and recycled for FY 2024-25 [N1]; FY 2025-26 data show 9,79,080 t recycled of 14,14,645 t generated, about 69% [N2]. The share has risen from about 10% in 2017-18 [N13], but roughly 30% remains unaccounted for on the official model
Uneven geography	Recycling is concentrated in Uttar Pradesh (3,76,896 t), Telangana (1,61,988 t) and Haryana (1,35,951 t) in FY 2025-26 [N2]; Maharashtra, Rajasthan and Punjab collect less relative to size [N13]
Data is modelled	Generation is estimated from sales and product life, not counted at disposal [N1]. Other estimates (1.6 Mt CPCB earlier, 4.1 Mt UN, per the first PDF) differ widely
Informal sector	Reported shares range from over 50% to 90-95% in studies cited in the first PDF; informal buyers pay more and carry no compliance cost
EPR can reward paperwork	A 2026 Delhi field study argues certificate markets monetise documentation rather than acquisition (first PDF)
Weak enforcement and leakage	Reports of fake certificates and authorised entities selling collected waste to unauthorised recyclers (first PDF); the November 2024 amendment now ties penalties to section 15 of the Environment (Protection) Act [N5, N6]
Institutions have no counting duty	The 2022 Rules removed bulk-consumer record keeping and annual returns (first PDF)
Data fear blocks disposal	Hoarding rises where people worry about data and lack a trusted channel
Limited reuse channels	Few structured refurbishing channels for institutions despite refurbishing certificates in the Rules
Repair culture still developing	Right to Repair framework and portal exist, but dedicated legislation is still discussed [N12]
Battery stream outpaces capacity	Formal lithium-ion recycling is small relative to expected retirements (first PDF); Battery Rules, 2022 now require EPR registration and set targets [N8]
Low awareness	Surveyed students show low knowledge of the policy and frequent dumping or storing
Poor recovery rates (older study)	A Bengaluru study reported gold recovery of 8-18% in Indian recyclers against over 95% internationally; check for newer data before quoting

PART 7. Disadvantages and risks of the national system

Disadvantage	Explanation	Mitigation
Cannot out-pay the informal market	A small payment may not attract high-value items, leaving the formal hubs with low-value costly waste	Compete on certificate, convenience, reuse; target institutions and companies; seek deposit-refund style funding
Data-breach liability	Holding devices before wiping makes the operator responsible. The DPDP Act and Rules were notified in November 2025; main duties apply from 14 May 2027 [N9]. The first PDF cites penalties up to Rs 250 crore for security failures and Rs 200 crore for failing to notify a breach (CONFIRM)	Secure cage, access logs, encryption, sealed bags, breach plan; legal advice on fiduciary or processor status
Cyber-incident duties	CERT-In requires specified incidents to be reported within 6 hours and logs kept 180 days in India [N10]	Indian hosting, runbook, named officer
Wiping is slow and imperfect	Large drives take hours; SSD and phone wiping depends on vendor support; locked devices cannot be wiped	Parallel stations, purge by IEEE 2883, destroy storage when in doubt
False sense of security	A certificate proves a process ran, not that no data remains	Sampled forensic checks and independent audit
Fraud and adverse selection	Payment can attract stolen, counterfeit or imported devices and worthless dumping	KYC, IMEI check, caps, photos, refuse unclear items
Cost and funding gap	Labour, tools, storage, insurance, audit and payouts need money; EPR income is uncertain	Phase-gated spending; recycler partnerships; resale margin; CSR
Legal and registration burden	E-Waste Rules 2022, Battery Rules, DPDP, GST, local approvals; a student hub is not a registered entity	Operate as a collection point feeding registered partners; written approvals
Vision cost and accuracy	Expensive, data hungry, cannot see inside devices	Use at recycler scale; intake demo only at hubs
Trade-in rebound	Paying for old devices can encourage faster replacement	Repair-first messages; price reuse above scrap
Livelihood impacts	Formalisation can displace informal collectors	Register them as aggregators or workers
Single point of failure	Central platform or hub can fail or be attacked	Backups, minimal data, manual fallback
Concentration of power	One national operator could dominate	Governance Council, independent audit, open price method
Environmental costs	Transport, wiping energy, destroyed storage	Consolidate trips, measure kWh, destroy only storage
Pilot may not generalise	A campus is not a country; students rotate	Document as a pilot; hand over to a standing club; gate each phase
Safety	Battery fires, sharps, dust, lifting	Training, PPE, fire-safe storage, no dismantling

Bottom line. The model addresses real causes (no trusted channel, data fear, no reuse, no independent visibility), but its economics and legal exposure are the hardest parts. Pilot without cash, prove collection and trust, measure, then add payment, company contracts and new regions with legal approval.

PART 8. Energy and environment calculations

- **Avoided emissions from reuse** = devices reused x emissions of making a comparable new device x displacement factor (the fraction of reused devices that truly replace a new purchase). The first PDF gives about 227-270 kg CO2 for making a laptop (ASU study) and says manufacturing and materials account for 70-90% of a smartphone's life-cycle emissions (industry body). Use a conservative factor such as 0.5 and state it.
- **Hub energy** = measured kWh per wipe x devices wiped x grid emission factor. Measure with a plug-in meter; the figure is small and should be reported honestly.
- **Transport emissions** = kilometres x vehicle emission factor. Use state lanes and consolidated trips.
- **Material recovery** = kg handed to registered recyclers x recovery fractions from the recycler's report. Do not claim more than the recycler documents.
- **Net benefit** = avoided emissions minus hub energy minus transport emissions, shown with assumptions and ranges.

Worked illustration (not a forecast)

Suppose a national phase handles 500,000 devices a year and 5% are laptops that are reused: 25,000 laptops. With 227 kg CO2 per new laptop and a displacement factor of 0.5, avoided emissions are 25,000 x 227 x 0.5 = 28,37,500 kg, about 2,838 tonnes CO2. At 270 kg the figure is about 3,375 tonnes. Subtract wipe energy and transport measured in the pilot. The result depends heavily on the reuse share and displacement factor, so report it as a range and never as a national claim.

National indicators to report

Indicator	Definition
Kg collected and recycled	Platform weights beside CPCB state figures, with the CPCB figures labelled as estimates
Platform recycling rate	Kg recycled divided by kg collected, from recycler records
Reuse rate	Devices graded A or B and reused divided by devices received
Wipe pass rate	Sampled recovery tests passed divided by tests run
Flag rate	Flagged devices divided by devices inspected
Cost per device and payout per kg	Operating cost and money paid divided by count and weight
Net emissions benefit	As calculated above with stated assumptions

PART 9. National implementation plan

Phase	Scope	Payment	Gate to move on (suggested [design])
0. Campus pilot (one semester)	Thapar campus: bins, booking, passport, intake, wipe demo, hold-and-escort of other devices, awareness, baseline and end surveys	Points or certificates	Written approvals; no data incident; every dispatch reconciled to recycler receipts; price survey done; trust rises in the second survey
1. City (months 5-12)	Patiala: institutions, offices, residents' associations; one registered recycler and one refurbisher under agreement	Cash after legal and tax sign-off	Weight-mismatch flags under 5%; wipe-sample failures under 1%; payout disputes under 1%; audit clean
2. State (year 2)	Punjab and the Chandigarh region; company contracts	Cash and points	Break-even on the B2B mix; second recycler partner; independent audit published
3. Region (years 3-4)	North India including Haryana, Delhi region and Uttar Pradesh, where most registered recycling sits [N2]	Cash and points	About 100,000 devices a year; incident drill passed; DPDP duties met
4. National (year 5 on)	Top generating cities and states; partner-run hubs on the same platform; Council in place	As approved	Council and audit functioning in every region; data shared with regulators on request

Pilot semester plan

Weeks	Activities	Output
1-2	Confirm topic, permissions, safety rules; draft the model	Approved scope
3-4	Audit and survey; price survey; select registered partners	Baseline data and quotes
5-7	Install bins; booking form and QR register; champions; awareness launch	Pilot live
8-10	Intake and triage; repair cafe; wipe demo; vision classifier training and testing	Demo-ready prototypes
11-12	Second survey; weigh outputs; energy and carbon sheet; reconcile with recycler receipts	Results
13	Record the video; slides; rehearse viva	Submission

Mapping to the 10-minute video

Segment	Time	Content	Presenters (7-8 members)
Problem and concern	1.5 min	National scale, official data, hoarding, data fear	1-2
Existing solutions	1.5 min	2022 Rules, EPR, informal sector, campus drives; limits (Part 6)	1
Idea	5 min	National governance model; pipeline; pricing; wipe demo; vision demo; awareness	3-4
Feasibility and applicability	2 min	Pilot results, risks (Part 7), phase gates	1-2

Course rules: real-time footage, each member presents a part, group and roll numbers at the start, no animation, no downloaded material, under 25 MB.

PART 10. Viva questions

Question	Short answer
Official data say about 70% is already recycled. Why is this needed?	That percentage is modelled from producer sales, and it does not show data-erasure trust, reuse, traceability to the device, or access for households and small institutions. The platform adds device-level evidence and independent reconciliation.
Why pay people for e-waste?	Informal buyers pay, and without a payoff or trust people hoard or sell informally. Payment is paired with certified erasure, convenience and reuse value.
Can you out-pay kabadiwalas?	Not on price alone; studies report informal buyers pay at least double. We compete on certificates, convenience, reuse and institutional demand, and test it with a price survey.
Is a factory reset enough?	No. NIST treats it as the lowest level (Clear). For flash storage we use the drive's secure-erase or cryptographic-erase commands, and destroy storage that cannot be sanitised.
What if the device is stolen?	We check the IMEI on Sanchar Saathi and refuse blocked or doubtful devices; owners sign a declaration.
Is computer vision necessary?	Not at hubs. It is valuable at the recycler's plant; we use it at intake to classify items and as a demonstration.
What are your legal risks?	Registration status of the hub, dealing only with registered refurbishers and recyclers under the 2022 Rules, DPDP Act duties from 14 May 2027, CERT-In incident duties, and battery safety. We operate as a collection point feeding registered partners.
How does it fit with EPR?	It does not issue EPR certificates. It gives registered partners a documented, audited inflow and reconciles weights, which supports traceability.
Who governs a national system?	A voluntary Governance Council with industry, consumer, informal-collector and academic members plus an independent auditor, working within the statutory framework and inviting CPCB as observer.
What is the main weakness?	Economics and legal exposure. We start with a no-cash pilot and gate each expansion.
How does it help the environment?	Reuse avoids new manufacturing, registered recycling avoids informal pollution and mining, awareness reduces waste; we quantify with a transparent method and a stated displacement factor.

Sources

Official sources first. Secondary sources (law-firm, news and compliance summaries) were used where the official text was not retrieved; confirm those points against the original. Statements marked 'first PDF' come from the uploaded document and its own source list (The India Forum 2026, Toxics Link, IIM Bangalore, ICRIER, US EPA, ASU laptop study and others); re-check them before quoting.

ID	Source	Used for
N1	PIB, Ministry of Environment answers in Parliament, PRID 2147876 and PRID 2155124, pib.gov.in	Generation 12,54,286 t and 13,97,955 t; 61.94% and 70.71% processed; method of estimation
N2	Ministry data reported by ETV Bharat, March 2026	FY 2025-26 generation and recycling; leading states
N3	CPCB FAQ on the E-Waste (Management) Rules, 2022, cpcb.gov.in and eprewaste.cpcb.gov.in; CPCB EPR portal	Registration of producers, recyclers, refurbishers; refurbisher consent to operate; 106 categories
N4	MoEFCC Rules and Regulations page, moef.gov.in/rules-regulations-3	Rules in force and amendments (2024)
N5	Lexplosion Solutions on the 2024 Amendment (secondary)	Dismantler definition; relaxation of return periods
N6	TeamLease RegTech on G.S.R. 699(E), 12 Nov 2024; Saikrishna and Associates on 2022 Rules (secondary)	Penalties under section 15; EPR targets of 60, 70, 80%
N7	Zerotrash and GreenSutra explainers (secondary)	Storage limit of 180 days; refurbishing certificates
N8	Battery Waste Management Rules, 2022 (MoEFCC), via Down To Earth and IEA (secondary)	Battery EPR, registration, informal collectors
N9	MeitY DPDP Rules, 2025 (notified 14 Nov 2025); phase dates per AMS Shardul (secondary)	DPDP timeline: 14 Nov 2025, 14 Nov 2026, 14 May 2027
N10	CERT-In Directions of 28 April 2022, cert-in.org.in; Mondaq and AZB summaries (secondary)	6-hour reporting; 180-day logs in India
N11	Department of Telecommunications press release, 16 May 2023, sancharsaathi.gov.in	CEIR and Know Your Mobile
N12	Department of Consumer Affairs Right to Repair portal (righttorepairindia.gov.in); IIM Bangalore note; Drishti IAS and Rau's IAS summaries (secondary)	Right to Repair framework, portal launched December 2022, sectors covered
N13	Dataful analysis of CPCB data (secondary)	10% to 70% trend; under-collecting states
N14	Uploaded governance model PDF, 12 October 2026	Structure, pipeline, flaws, pricing formula, standards table, calculations, viva

Re-check every figure and legal point against the original source before quoting it. Laws, portals and prices change.